

Part A: Review of Week 2 Analysis & Gap Identification

Issue / Gap Identified	Evidence from Week 2 Analysis	Required Week 3 Improvement
1. Aggregated Channel & Fraud Analysis	Week 2 evaluated gross transaction volumes per channel without segmenting by risk status or success rate.	Conduct multi-dimensional SQL CTEs and cohort analysis joining transaction status with synthetic Risk_Review_Flag across channels.
2. Superficially Defined Customer Lifetime Value & Recency	Customer segmentation relied solely on static demographic fields without behavioral RFM (Recency, Frequency, Monetary) metrics.	Implement SQL window functions (NTILE, DENSE_RANK) and Python RFM clustering to dynamically group users.
3. Static Visualization & Uncalibrated KPIs	Week 2 Power BI visualizations lacked cross-filtering logic, multi-level drill-downs, and explicit DAX measures for dynamic time-intelligence framing.	Re-architect the data model in Power BI into a Star Schema; construct optimized DAX measures for MTD/YTD growth, risk ratios, and channel efficiencies.